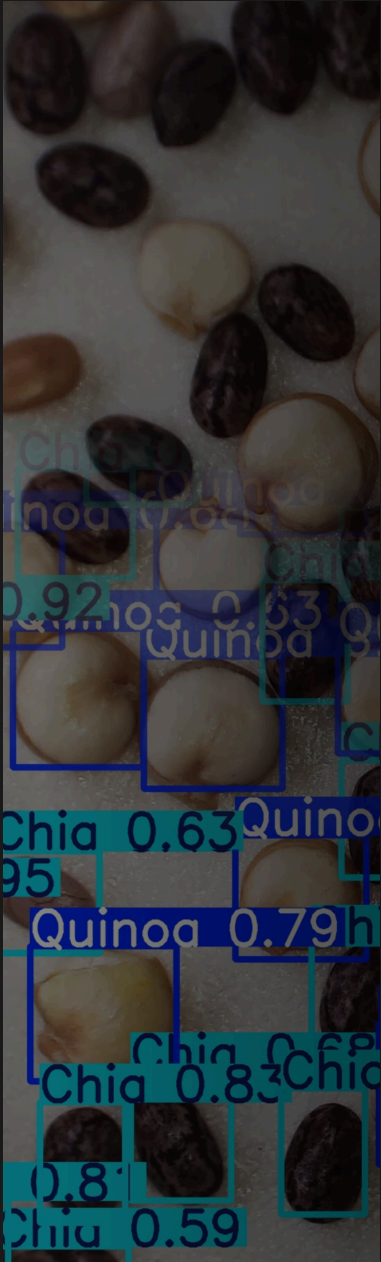




HANDS-ON COMPUTER VISION ON MICROSCOPES

Workshop Team (*EPFL Center for Imaging*):

Mallory Wittwer (Presenter)

Florian Aymanns

Jules Vanaret

Edward Andò

Daniel Sage

LET'S BUILD A SMART MICROSCOPE

EPFL  Center for Imaging

Configuration

Confidence Threshold
0.25
0.00 1.00

IoU Threshold
0.45
0.00 1.00

Classes

Quinoa seed x

Chia seed x

Start Camera

Object Detection with YOLO



Quinoa seed 0.68
Quinoa seed 0.65
Chia seed 0.81
Quinoa seed 0.55
Quinoa seed 0.72
Chia seed 0.74
Chia seed 0.40
Quinoa seed 0.47
Quinoa seed 0.59
Quinoa seed 0.38
Quinoa seed 0.61
Quinoa seed 0.60
Chia seed 0.76
Quinoa seed 0.56
Quinoa seed 0.55
Chia seed 0.64
Quinoa seed 0.50
Quinoa seed 0.50

Stop Camera

EPFL Center for Imaging

Configuration

Confidence Threshold

0.25
0.00 1.00

IoU Threshold

0.45
0.00 1.00

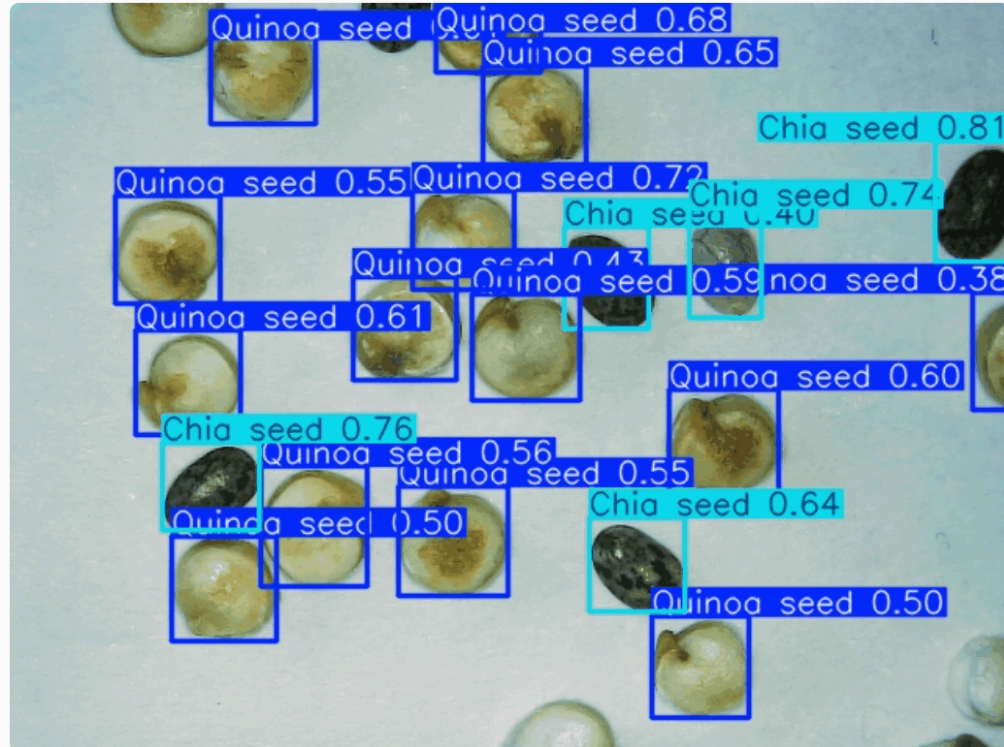
Classes

Quinoa seed x

Chia seed x

Start Camera

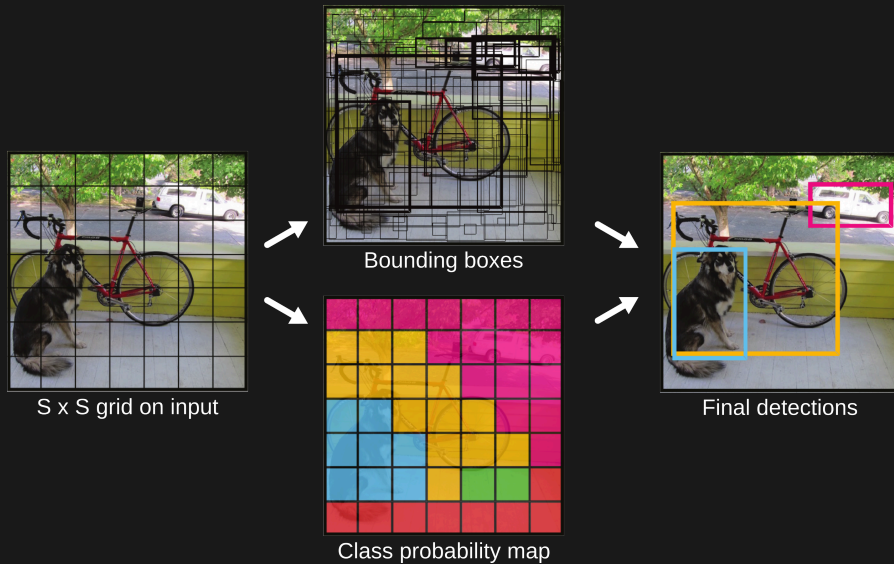
Object Detection with YOLO



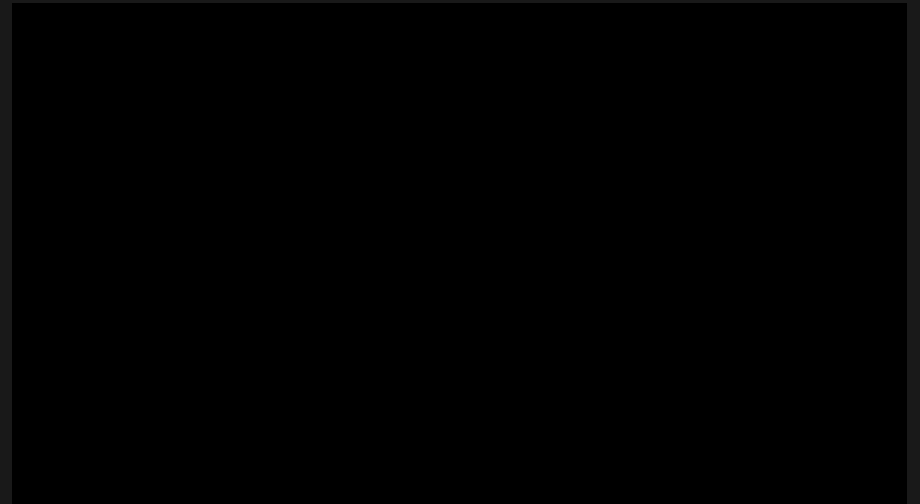
Stop Camera

YOLO (YOU ONLY LOOK ONCE)

Detect bounding boxes and **classify** them in one go.



Original YOLO detector (2015)



YOLO11 (2025)

Source: Ultralytics

YOLO (YOU ONLY LOOK ONCE)

- Optimized for **real-time applications**.
- We'll use the implementation from the [Ultralytics](#) library in Python.
 - Simple, well-maintained, consistent API (*train, predict*, etc.).
 - Provides **pretrained models, data augmentations, and sensible defaults**.
 - Licensed under [AGPL-3.0](#).

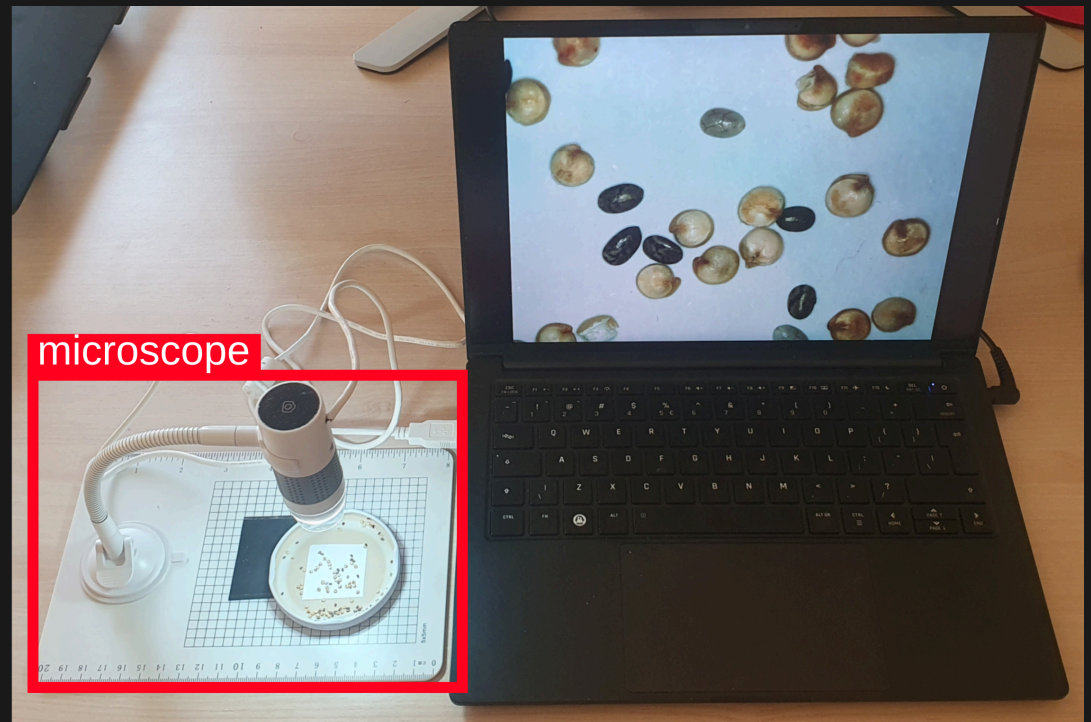
YOUR IMAGING EQUIPMENT FOR TODAY

USB microscopes

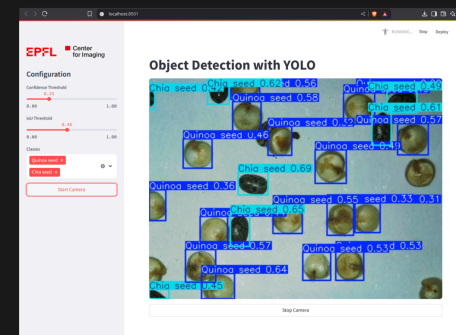
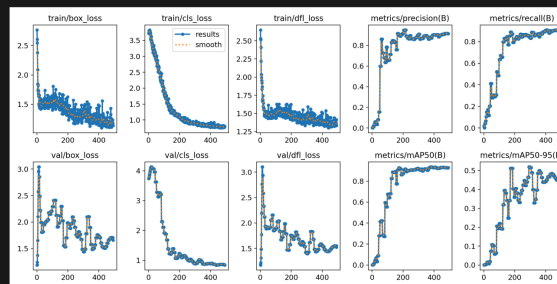
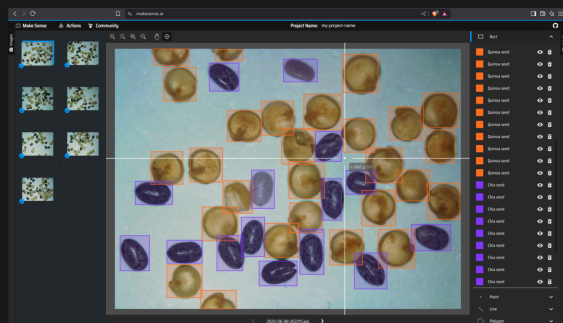
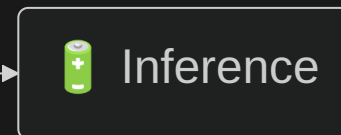
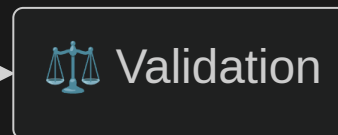
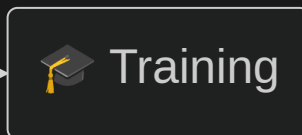
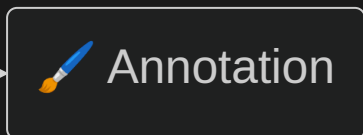
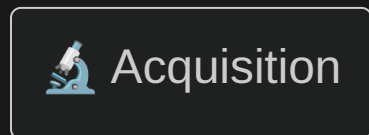
Magnification: 60-200 X

Connection: USB-A

Cost: ~70 CHF



OVERVIEW OF THE STEPS INVOLVED



WHAT WE HOPE YOU'LL TAKE AWAY WITH YOU

-  Not the microscopes; please return them at the end of the session! :)
-  Understand the main steps of a computer vision project through practice.
-  Get new ideas and insights that you can apply to your own projects.

YOUR TURN



UNTIL 15:30.

Make groups of 3



You should be equipped with:



1X USB microscope



Laptop with Python + USB-A



Some dry seeds



Workshop guide



<https://go.epfl.ch/yolo-workshop>